

MAT 426, Class 7 Homework: Orthogonality
(Due Tuesday, March 3, 2026)

1. Let $\mathbf{u} = \begin{bmatrix} -3 \\ 7 \\ 2 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} -1 \\ 5 \\ 3 \end{bmatrix}$, and $\mathbf{w} = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}$. Compute the following quantities.

(a) $\frac{1}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}$

(b) $\mathbf{u} \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{w}$

(c) $\frac{\mathbf{u} \cdot \mathbf{w}}{\mathbf{w} \cdot \mathbf{w}}$

(d) $\|\mathbf{u}\|$

2. Find a unit vector in the direction of $\mathbf{u} = \begin{bmatrix} -4 \\ 1 \\ 5 \end{bmatrix}$.

3. Find the distance between $\mathbf{x} = \begin{bmatrix} 10 \\ -3 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} -1 \\ 5 \end{bmatrix}$.

4. Determine which pairs of vectors are orthogonal.

(a) $\mathbf{a} = \begin{bmatrix} 8 \\ -5 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$

(b) $\mathbf{u} = \begin{bmatrix} 12 \\ 3 \\ -5 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 2 \\ -3 \\ 3 \end{bmatrix}$

(c) $\mathbf{u} = \begin{bmatrix} 3 \\ 2 \\ -5 \\ 0 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} -4 \\ 1 \\ -2 \\ 6 \end{bmatrix}$

(d) $\mathbf{y} = \begin{bmatrix} -3 \\ 7 \\ 4 \\ 0 \end{bmatrix}$, $\mathbf{z} = \begin{bmatrix} 1 \\ -8 \\ 15 \\ -7 \end{bmatrix}$

5. Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$. Explain why $\mathbf{x} \cdot \mathbf{x} \geq 0$. When is $\mathbf{x} \cdot \mathbf{x} = 0$?

6. Suppose $\mathbf{u}, \mathbf{v}$ are vectors in $\mathbb{R}^n$. Prove that $\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$.

7. Let $\mathbf{u} = \begin{bmatrix} a \\ b \end{bmatrix}$. Describe the set of vectors $\begin{bmatrix} x \\ y \end{bmatrix}$ that are orthogonal to $\mathbf{u}$. *Hint:* Consider $\mathbf{u} = \mathbf{0}$ and $\mathbf{u} \neq \mathbf{0}$.

8. Suppose a vector $\mathbf{y}$ is orthogonal to $\mathbf{u}$ and $\mathbf{v}$. Prove that $\mathbf{y}$ is orthogonal to the vector $\mathbf{u} + \mathbf{v}$.

9. Suppose that $\mathbf{y}$ is orthogonal to $\mathbf{u}$ and $\mathbf{v}$. Prove that $\mathbf{y}$ is orthogonal to every $\mathbf{w}$ in $\text{Span}\{\mathbf{u}, \mathbf{v}\}$.

10. Let W be a subspace of $\mathbb{R}^n$ and let $W^\perp$ be the set of all vectors orthogonal to W .
- (a) Show that $\mathbf{0}$ is an element of $W^\perp$.
 - (b) Let $\mathbf{z} \in W^\perp$ and c be a scalar. Let $\mathbf{u}$ be an arbitrary element of W . Show that $c\mathbf{z}$ is orthogonal to $\mathbf{u}$.
 - (c) Let $\mathbf{z}_1, \mathbf{z}_2 \in W^\perp$. Let $\mathbf{u}$ be an arbitrary element of W . Show that $\mathbf{z}_1 + \mathbf{z}_2$ is orthogonal to $\mathbf{u}$.
 - (d) What do (a)-(c) tell you about $W^\perp$?
11. Show that if $\mathbf{x}$ is in both W and $W^\perp$, then $\mathbf{x} = \mathbf{0}$.